CITY GENERAL HOSPITAL

Department of Pathology & Laboratory Medicine
Arterial Blood Gas Analysis Report

Patient ID: ABG-TEST-001	**Ward / Unit:** Post-Op Recovery (ICU-4)
Patient Name: Test Patient	**Requesting Clinician:** Dr. J. Smith
DOB: 1975-03-14 (Age: 50)	**Collection Time:** 2026-04-15 09:42
MRN: MRN-2026-00541	**Report Time:** 2026-04-15 09:55

CLINICAL CONTEXT

Post-operative day 2 following emergency laparotomy for bowel perforation. Patient currently on 40% O2 via Venturi mask (FiO2: 0.40). No known COPD. No ketosis. Haemodynamically borderline — MAP 62 mmHg.

ARTERIAL BLOOD GAS RESULTS

Parameter	Result	Unit	Reference	Flag
pH	7.28		7.35 – 7.45	L
PaO2	68	mmHg	80 – 100	L
PaCO2	52	mmHg	35 – 45	H
HCO3-	22	mmol/L	22 – 26	
Base Excess (BE)	-4.2	mmol/L	-2 – +2	L
Lactate	3.1	mmol/L	0.5 – 2.0	H
SpO2 (calc.)	91	%	95 – 100	L
FiO2	0.40	(fraction)	0.21 – 1.00	

ELECTROLYTES & METABOLIC PANEL

Parameter	Result	Unit	Reference	Flag
Na+	138	mmol/L	135 – 145	
Cl-	102	mmol/L	98 – 106	
Albumin	3.5	g/dL	3.5 – 5.0	
Glucose	6.2	mmol/L	3.9 – 6.1	H

PRELIMINARY INTERPRETATION

Acid-Base: Respiratory acidosis (pH 7.28, PaCO2 52 mmHg) with mild metabolic compensation (HCO3- 22 mmol/L). Base deficit -4.2 mmol/L consistent with mild metabolic acidosis component.

Oxygenation: Type II Respiratory Failure — hypoxia (PaO2 68 mmHg on FiO2 0.40, P/F ratio = 170) combined with hypercapnia (PaCO2 52 mmHg). SpO2 91% subtherapeutic.

Perfusion: Elevated lactate 3.1 mmol/L — consistent with tissue hypoperfusion. Clinical picture concerning for early sepsis in post-operative setting.

Electrolytes: Na 138, Cl 102 — anion gap borderline elevated. Albumin 3.5 g/dL. Mildly elevated glucose 6.2 mmol/L, no ketosis identified.

LABORATORY USE ONLY — For clinical decision support. Verified by: Auto-analyser ABL90 FLEX PLUS. This report is confidential and intended solely for the requesting clinician.